



Watching the Wolves: Unveiling the Moderating Role of Corporate Governance on CEO Power

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Introduction & Motivation: Powerful CEOs Matter

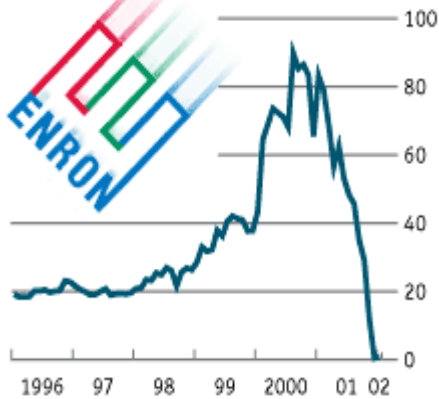
- Prior studies suggest that CEO power has a negative effect on corporate outcomes
 - Powerful CEOs may influence the board to extract
 - » High compensation (Morse, Nanda, Seru, 2011; Bebchuk, Cremers and Peyer, 2011)
 - » Private benefits from more and less valuable M&A deals (Grinstein and Hribar, 2004)
 - » Adversely affect firm's governance and performance (Han et al., 2016; Adams et al., 2005; Khanna et al., 2015)
 - » Potentially exert sub-optimal levels of effort (Bertrand and Mullainathan, 2003)
- Key question?
 - Are there any governance mechanisms that may channel the misaligned efforts of powerful CEOs to value-enhancing projects of the firm?
- We focus on the role of independent directors in doing so.

Can Board Governance Discipline Powerful CEOs?

- *Board of directors* is a key mechanism of corporate governance to discipline the behaviour of self-serving managers (Adams, Hermalin and Weisbach, 2010)
- Firm governance structures are endogenous
 - Powerful CEOs may select board of directors to ensure a pliable board (Hermalin and Weisbach, 1998 , Tirole, 2001; Fama, 1980; Shivdasani and Yermack, 1999)

A star has fallen

Enron share price, \$



AP Photo / Michael Stravato



2013 Harris Poll RQ® Study Top 5 Corporate Reputations

1	amazon.com	82.62
2	Apple	82.54
3	The Walt Disney Company	82.12
4	Google	81.32
5	Johnson & Johnson	80.95

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Endogeneity of Board Governance

- Mechanism: Concurrent passage of the Sarbanes-Oxley Act and NYSE/NASDAQ listing regulation changes (collectively, “SOX”) as a quasi-exogenous natural experiment
- Why SOX?
 - Graham, Kim, and Leary (2017) report that average board independence increased to over 70% by 2011 following the introduction of SOX and amendments to NYSE and NASDAQ listing requirements
 - Guo and Masulis (2015) assert that SOX enactment is a promising way to overcome the endogeneity of board monitoring
 - Banerjee et al (2015) show that SOX helps to reign in overconfident CEOs

Was SOX Enactment Quasi-natural for All Firms?

- Complicating factor: The empirical evidence on the effect of SOX on corporate policies is inconclusive (Coates and Srinivasan, 2014)
 - Why? Failure to explicitly consider the heterogeneity in underlying firm governance mechanisms at the time of enactment of these regulations
- Firms with a majority independent board and fully independent audit committee before it became mandated are arguably better governed than the pre-SOX *Non-Compliant Firms*
- SOX is unlikely to be an “exogenous shock” for pre-SOX Compliant firms
- Our study focuses on pre-SOX Non-Complaint firms

Sample Design and Methodology

- Sample: S&P1500 firms during 1992-2014
 - After excluding firms that experience turnover of CEOs around the SOX in 2002, we have 1,102 unique firms with 2,909 CEOs and 18,396 firm-year observations
 - CEO power: Founder, CEO-Chair duality, Title concentration, Tenure above industry median, Ownership above industry median
 - CEO Power TOP-Q: one if 'CEO Power' index is in the top quartile of the industry

- Baseline Specification

$$Y_{i,t+1} = \alpha + \beta SOX_{i,t} \times CEOPower_{i,t} + \partial CEOPower_{i,t} + \delta SOX_{i,t} + \omega \gamma_{i,t} + \phi_t + \lambda_{j(i)} + \varepsilon_{i,t}$$

- Fixed effects: Year Fixed effects , firm (Industry) Fixed effects
- Control Variables: Firm Size, Profitability, CAPEX, R&D & Tobin's Q among others.

Basic Research Question and Key Findings

- Does the exogenous improvement in board governance after SOX encourage powerful CEOs to adopt optimal corporate policies?
- Findings:
 - Exploring the heterogeneity in a firm's pre-SOX governance, we show that
 - » The empowered board can divert the energy and effort of self-serving powerful CEOs to value enhancing projects and initiate a strategic shift in the firms without necessarily curbing CEO power
 - Our study supports the views:

“Power [is] the basic energy to initiate and sustain action translating intention into reality, the quality without which leaders cannot lead.” – Bennis and Nanus (1985)
 - A powerful CEO coupled with poor corporate governance drives the negative views of powerful CEOs

H1: SOX increases investment in value enhancing R&D projects by in Non-Compliant Firms with powerful CEOs.

Dependent variable	[RD(t+1)/Asset(t)] x 100			
	Non-compliant		Non-compliant	
Model	(1)	(2)	(3)	(4)
SOX * CEO Power	0.123*** [0.005]	0.159*** [0.002]		
SOX *CEO power Top Q			0.253** [0.023]	0.422*** [0.005]
CEO Power	-0.144*** [0.000]	-0.135*** [0.000]		
CEO power Top Q			-0.324*** [0.006]	-0.247* [0.061]
SOX	0.439* [0.061]	0.869*** [0.001]	0.668*** [0.004]	1.146*** [0.000]
Year fixed effect	Y	Y	Y	Y
Firm fixed effect	N	Y	N	Y
Industry fixed effect	Y	N	Y	N

- Powerful CEOs might derive more disutility from R&D investments as they value control over larger resources and failed investments in R&D may dissipate those resources.

Dependent variable	Patent $(t+2)$	Citations $(t+2)$	Log(Market value of innovation)	Patent $(t+2)$	Citations $(t+2)$	Log(Market value of innovation)
Model	(1)	(2)	(3)	(4)	(5)	(6)
SOX * CEO Power	0.048** [0.044]	0.132*** [0.003]	0.085** [0.047]			
SOX * CEO power Top Q				0.316*** [0.000]	0.525*** [0.001]	0.508*** [0.001]
CEO Power	-0.043** [0.012]	-0.089** [0.010]	-0.066* [0.083]			
CEO power Top Q				-0.102 [0.108]	-0.168 [0.209]	-0.358*** [0.004]
SOX	-1.230*** [0.000]	-2.426*** [0.000]	1.375** [0.042]	-1.171*** [0.000]	-2.225*** [0.000]	1.511** [0.018]
Firm size	-0.027 [0.261]	0.007 [0.859]	0.185*** [0.000]	-0.025 [0.299]	0.009 [0.805]	0.183*** [0.000]
Profitability	-0.311 [0.223]	0.312 [0.387]	-0.626 [0.183]	-0.300 [0.246]	0.343 [0.339]	-0.643 [0.166]
CAPEX/Asset	0.155 [0.274]	0.120 [0.728]	0.662*** [0.000]	0.147 [0.303]	0.104 [0.766]	0.705*** [0.000]
R&D/Asset	3.490*** [0.003]	11.267*** [0.000]	-3.257*** [0.000]	3.409*** [0.004]	11.129*** [0.000]	-3.310*** [0.000]
Tobin's Q	0.063 [0.111]	0.014 [0.825]	-0.001 [0.972]	0.058 [0.136]	0.008 [0.903]	0.003 [0.913]
Constant	1.233*** [0.000]	2.356*** [0.000]	2.133*** [0.000]	1.154*** [0.000]	2.188*** [0.000]	2.179*** [0.000]
Year fixed effect	Y	Y	Y	Y	Y	Y
Firm fixed effect	Y	Y	Y	Y	Y	Y
Observations	5,055	5,055	2,542	5,055	5,055	2,542
R-squared	0.285	0.379	0.422	0.288	0.379	0.426

- After SOX, powerful CEOs' innovations are more valuable than before, as measured by
 - Patents
 - Citations
 - Value of innovation (per KPSS, 2017 database)
- **Why?**
 - Timely feedback to managers by empowered board in the post-SOX (Manso, 2011).
 - Intensified monitoring coupled with diverse expert opinions from the restructured board (Balsmeier et al., 2017).

H2: SOX reduces over-investment in CAPEX and PP&E by Non-Compliant Firms with powerful CEOs.

Dependent variable	CAPX Growth			PPE Growth		
Model	(1)	(2)	(3)	(4)	(5)	(6)
SOX * CEO Power	-0.020*** [0.009]	-0.025*** [0.006]		-0.009** [0.025]	-0.007** [0.046]	
SOX *CEO power Top Q			-0.058** [0.042]			-0.031** [0.046]
CEO Power	0.016*** [0.004]	0.012 [0.103]		0.012*** [0.000]	0.008** [0.014]	
CEO power Top Q			0.036 [0.180]			0.018 [0.230]
SOX	0.534*** [0.000]	0.490*** [0.000]	-0.306*** [0.000]	-0.106*** [0.000]	-0.051*** [0.000]	-0.113*** [0.000]
Firm level controls	Y	Y	Y	Y	Y	Y
Year fixed effect	Y	Y	Y	Y	Y	Y
Industry fixed effect	Y	N	N	Y	N	N
Firm fixed effect	Y	Y	Y	Y	Y	Y

- *Non-Compliant Firms* with powerful CEOs engage
 - Less in empire building (pursue lower growth in capital expenditures (CAPEX), and property, plant, and equipment (PPE)) and thus reduce physical investments in the post-SOX period
 - Undertake higher quality investments (positive announcement returns on M&A deals) but fewer acquisitions (measured by the number of deals) in the post-SOX period.

Value creation in product announcements

Dependent variable	75 th percentile CAR		CAR	
Model	(1)	(2)	(3)	(4)
SOX * CEO Power	0.271*** [0.009]		0.018** [0.046]	
SOX * CEO power Top Q		0.529** [0.042]		0.079* [0.065]
CEO Power	-0.188** [0.026]		-0.017** [0.034]	
CEO power Top Q		-0.478** [0.048]		-0.089* [0.077]
SOX	1.315* [0.080]	1.675** [0.027]	0.138** [0.018]	0.124* [0.076]
Firm size	0.233 [0.668]	-0.208 [0.554]	0.009 [0.834]	-0.001 [0.980]
Sales growth	-0.375** [0.034]	-0.311** [0.025]	-0.022* [0.085]	-0.023* [0.086]
Profitability	-1.300 [0.102]	-1.336 [0.332]	-0.106 [0.121]	-0.148 [0.336]
CAPX/Asset	-6.955 [0.247]	-2.094 [0.545]	-0.646 [0.139]	-0.689 [0.164]
R&D/Asset	6.675** [0.040]	5.909 [0.220]	0.482** [0.035]	0.480 [0.233]
Tobin's Q	0.176*** [0.000]	0.097 [0.139]	0.016*** [0.000]	0.014 [0.210]
Patents(t)	-0.067 [0.572]	0.016 [0.834]	-0.010 [0.294]	-0.011 [0.254]
Constant	-4.320 [0.375]	-0.062 [0.985]	-0.091 [0.801]	-0.002 [0.997]
Year fixed effect	Y	Y	Y	Y
Firm fixed effect	Y	Y	Y	Y
Observations	1,381	1,264	1,381	1,264
R-squared	0.171	0.174	0.205	0.172

- But do powerful CEOs now create more value when they do invest?
- Returns to new product announcements are higher after SOX
- Economically meaningful
 - After SOX, powerful CEOs' product announcements generate 1.8% higher event returns than before
- Note: product announcement data from Mukherjee et al (2016)

What about acquisitions

Dependent variable Models	CAR(-2,2)		Acquisition count	
	(1)	(2)	(3)	(4)
SOX * CEO Power	0.006** [0.032]		-0.076** [0.028]	
SOX *CEO power Top Q		0.020** [0.029]		-0.262** [0.043]
CEO Power	-0.004* [0.070]		0.029 [0.255]	
CEO power Top Q		-0.010 [0.229]		0.076 [0.380]
SOX	0.044** [0.023]	-0.031** [0.028]	-0.225 [0.245]	-0.391** [0.030]
Relative deal size	-0.014 [0.233]	-0.015 [0.412]	-0.441*** [0.000]	-0.436*** [0.000]
Subsidiary target	0.001 [0.770]	0.001 [0.722]	-0.081 [0.659]	-0.070 [0.704]
Cross boarder deal	0.002 [0.716]	0.001 [0.574]	-0.078** [0.039]	-0.077** [0.039]
Serial bidder	0.006 [0.223]	0.006* [0.096]	-0.049 [0.335]	-0.051 [0.320]
Constant	0.103** [0.015]	0.097* [0.055]	-0.960** [0.025]	-0.896** [0.037]
Firm level controls	Y	Y	Y	Y
Year fixed effect	Y	Y	Y	Y
Firm fixed effect	Y	Y	N	N
Observations	1,905	1,905	1,905	1,905
R-squared	0.035	0.033		

- Acquisitions are a major flashpoint for agency conflicts (see e.g., Masulis et al, 2007; Harford et al, 2012)
- Acquisition CARs increase between 0.6 and 2 percentage points after SOX for powerful CEOs
- Seems to be at least a function of them doing fewer acquisitions, suggesting they are cutting some of the more marginal deals

H3: SOX increases dividend payments by Non-Compliant Firms with Powerful CEOs.

Dependent variable	Dividend _(t+1) x100	
Model	(1)	(2)
SOX * CEO Power	0.098** [0.042]	
SOX *CEO power Top Q		0.517*** [0.008]
CEO Power	-0.086** [0.021]	
CEO power Top Q		-0.406** [0.010]
SOX	-0.092 [0.643]	0.058 [0.740]
Firm level controls	Y	Y
Year fixed effect	Y	Y
Firm fixed effect	Y	Y

- Powerful CEOs who might have moral hazard problem could further induce agency conflicts as the availability of free cash flow resulting from a firm's payout policy may lead to engagement in wasteful investment or extraction of perks.
- Dividend signalling models argue that more valuable firms pay higher dividends (Bhattacharya, 1979).

Conclusions:

- We use the concurrent passage of the Sarbanes-Oxley Act (SOX) of 2002 and NYSE/NASDAQ listing regulation changes as an exogenous shock to internal firm governance to explore the impact of powerful CEOs on corporate policies.
- We use the heterogeneity in a firm's pre-SOX governance, to challenge the notion that CEO power negatively impacts corporate outcomes for all firms.
- We show that for firms with weaker pre-SOX corporate governance (Non-Compliant Firms) and powerful CEOs, the improvement in governance induced by exogenous regulatory changes initiated a strategic shift in resource allocation.
- As falsification test, we show that moderating effect of corporate governance through SOX on CEO power is not consistent for the pre-SOX Complaint Firms and less powerful CEOs (bottom decile of power index)
- As additional test, we report that the SOX had stronger discipline effect on the pre-SOX Non-complaint firms with entrenched powerful CEOs.

Conclusions

- Our results are robust to controlling for
 - Other plausible measures of CEO Power
 - » CEO pay slice (CPS, BCF 2009)
 - » CEO being the only insider in the board
 - Other CEO characteristics
 - » CEO overconfidence
 - Other governance mechanisms
 - » Institutional holdings, Board size, Dual class-stock structure
 - Other firm-level characteristics-Leverage, Cash-flows
 - Use of high-dimensional fixed effects
 - Triple interactions analysis including the compliant firms
- Our results are not driven by any particular source of power solely

Summary statistics

Panel A: CEO Power components and variables			
Variable	Mean	Median	Standard deviation
Founder CEO	0.18	0	0.38
CEO-Chair duality	0.60	1	0.49
Title Concentration	0.25	0	0.44
CEO Ownership	0.03	0	0.06
CEO Ownership above industry median	0.47	0	0.5
CEO Tenure	8.86	6	7.96
CEO Tenure above industry median	0.50	0	0.50
CEO Power	2.02	2	1.40
CEO Power Top Q	0.16	0	0.37

Summary statistics

Variables	Panel D: Non-Compliant firms in pre-SOX period					
	Powerful CEO: Treated		Non-Powerful CEO: Control		Test of differences	
	Mean	Median	Mean	Median	t-stat	z-stat
Market Cap (\$million)	5433.09	895.40	9127.15	1292.01	1.71	3.72
Profitability	0.12	0.12	0.12	0.11	-0.85	-2.62
R&D/Asset	0.03	0.00	0.03	0.00	0.32	2.42
Patents _{t+2}	1.05	0.00	1.37	0.69	4.01	4.07
Citations _{t+2}	1.96	0.00	2.28	0.00	2.52	3.14
CAPEX/Asset	0.08	0.05	0.07	0.06	-2.42	0.87
Dividend/Asset	0.01	0.00	0.02	0.01	4.71	12.22
PPE Growth	0.14	0.12	0.11	0.06	-3.17	-5.48
Tobin's Q	2.82	2.02	2.38	1.76	-3.76	-5.16
Log(Market value of innovation)	3.59	3.33	4.29	4.05	4.05	4.20
CAR(-2,2)	0.61	0.56	0.26	0.22	-0.57	-1.02